

HLPW-5: Overview and Workshop Summary

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The Fifth AIAA CFD High-Lift Prediction Workshop was held with the goal of assessing the numerical prediction capability of current-generation computational fluid dynamics (CFD) technology for swept, medium/high-aspect-ratio wings in high-lift configurations. A key aspect of this endeavor was the use of Technology Focus Groups (TFG), which included both mesh-generation and flow-solver experts working together to accelerate advancements for their particular CFD methodology, by addressing key questions of importance prior to the workshop. The high-lift version of the NASA Common Research Model (CRM-HL) configuration was the focus of this workshop, and was used for three unique test cases. Wind-tunnel data were available for comparison for one of the test cases. Altogether, 365 datasets of CFD results were submitted by 47 teams, with 41 teams contributing to the multiple configurations of Case 1, 40 to Case 2, and 18 to Case 3. This paper provides a high-level summary of the results and conclusions from the workshop. As concluded from past workshops, fixed-grid Reynolds-averaged Navier-Stokes methods continued to be inaccurate and inconsistent for high-lift flows near maximum lift. However, application of mesh-adaptation technology helped to achieve improved consistency. Scale-resolving methods appeared most promising for predicting high-lift flow physics, particularly at maximum lift. Best practices for these methods were refined over the course of the workshop and new challenges were identified.

I. Nomenclature

C_L	=	lift coefficient
$C_{L,max}$	=	maximum lift coefficient
C_D	=	drag coefficient
C_M	=	moment coefficient
C_p	=	pressure coefficient
τ_w	=	wall stress
α	=	angle of attack
ρ	=	density
T	=	temperature
Re	=	Reynolds number
M	=	Mach number
MAC	=	Mean Aerodynamic Chord

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η = semispan fraction

II. Introduction

THE High-Lift Prediction Workshop (HLPW) series has been ongoing since 2010. Its goals include: (1) assess the numerical prediction capability (meshing, numerics, turbulence modeling, high-performance computing requirements, etc.) of current-generation CFD technology/codes for swept, medium-to-high-aspect ratio wings for landing/take-off (high-lift) configurations; (2) develop practical modeling guidelines for CFD prediction of high-lift flow fields; (3) determine the elements of high-lift flow physics that are critical for modeling to enable the development of more accurate prediction methods and tools; and (4) enhance CFD prediction capability for practical high-lift aerodynamic design and optimization. The fifth workshop, HLPW-5, was held in Las Vegas in summer 2024, associated with the AIAA Aviation Forum.

Previous workshop summaries can be found in their respective summary papers [1–9]. Many of these papers include references to other individual participant CFD studies as well. A high-level overview of the workshop series is also available [10]. Generally speaking, over the nearly 15 years since the first workshop in 2010, there has been a significant increase in available computational power, allowing most participants to run on significantly larger CFD grids than were previously possible. Because of this, improved grid-generation codes, and more robust CFD algorithms/capabilities, it has become easier to include all the geometry components (such as slat and flap brackets) of very complex high-lift configurations. Despite these improvements, CFD solutions based on the Reynolds-averaged Navier-Stokes (RANS) equations have failed as a whole to accurately and consistently predict high-lift characteristics near $C_{L,max}$, as compared to experimental observations. Mostly, this is attributed to the inability of RANS turbulence models to accurately model the flow physics associated with highly separated flows.

Starting around the time of HLPW-3 in 2017, it was recognized that scale-resolving simulation (SRS) methods had the *potential* to predict high-lift flowfields more accurately than RANS technology. In HLPW-4 two of the five Technology Focus Groups (TFGs) were organized to investigate both Hybrid RANS/Large-Eddy Simulation (HRLES) methods and Wall-Modeled LES (WMLES) methods, respectively. In that workshop, the results obtained in the HRLES and WMLES TFGs were reasonably successful at reproducing flowfields near $C_{L,max}$, generally agreeing better with experimental data than RANS results. Efforts in these two TFGs have continued into the current workshop to further explore, develop, and document SRS capabilities and best practices to predict high-lift flowfields.

The other three TFGs from HLPW-4, fixed-grid RANS, mesh adaptation (ADAPT), and high-order discretization (HO), also actively participated and contributed to HLPW-5. However, the HO TFG did not directly participate in flow solver studies, but instead focused primarily on the development of high-order meshes.

HLPW-5 was made up of three separate test cases of varying complexity. Each test case was evaluated sequentially by all TFGs together, with approximately six months dedicated to each individual case, and a mini-workshop conducted at the conclusion of test cases 1 and 2 to collectively review results. The test cases were designed such that lessons learned were nominally directly applicable to the future cases, and the mini-workshops enabled collaborative learning as the workshop progressed. This approach allowed participants to focus, together, on specific issues pertinent to one test case at a time, and greatly enhanced collaboration compared to previous workshops, where participants either worked in isolation or worked across a larger volume of test cases simultaneously.

The current paper documents the major high-level lessons learned from HLPW-5 across all of the TFGs. Companion papers from each TFG provide more technical details and results [11–15]. This paper is organized as follows. Section III provides a description of the test cases, Section IV summarizes the CFD meshes, and Section V summarizes the entries. Then, Section VI describes results, Section VII highlights the key takeaways, and the paper concludes with Section VIII.

III. Description of Test Cases

There were three unique test cases that were considered for HLPW-5, which each utilized different configurations of the High-Lift Common Research Model (CRM-HL) [16–18]. The CRM-HL is an open-source, publicly available commercial transport aircraft geometry representative of an industrial high-lift configuration, utilized for CFD validation within a broad international CRM-HL Ecosystem [19]. Specific test case geometry for this workshop either reflects the CRM-HL reference configuration, or explicitly matches the “as-designed” geometric CAD definition of one particular wind-tunnel model that was designed, built, and tested by an Ecosystem partner. The CRM-HL geometry was provided to HLPW-5 participants in full-scale inches, for which the MAC is 275.8 inches. In this paper, any mention of X, Y, or Z coordinates is in this same reference frame.

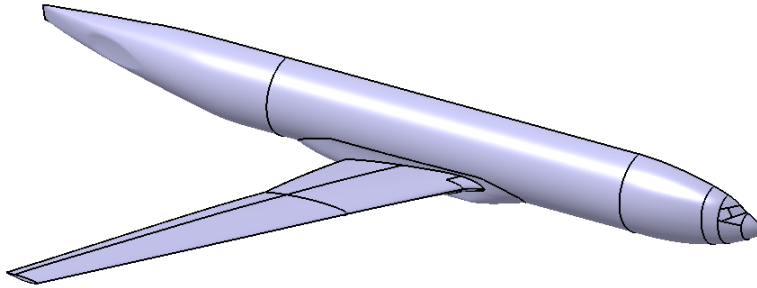
Table 1 Test Case 1 Parameters

Domain	Rectangular Cuboid
Domain Size	X: -65,000 to 65,000 in.
	Y: 0 to 65,000 in.
	Z: -65,000 to 65,000 in.
Mach	0.20
Chord Reynolds Number	5.6×10^6
α	11°
Reference Static Temperature	521 °R

The three test cases chosen for HLPW-5 were developed and organized to increase complexity sequentially, and were intended to help answer specific key questions aimed at advancing the goals of the workshop. Each test case and its specific objectives are described in the sections below.

A. Test Case 1: CRM-HL Wing-Body Verification

Previous High-Lift Prediction Workshops have highlighted the importance of solver verification to obtain consistent solutions for complex flow problems. For this workshop, the verification problem was chosen based on the simplest CRM-HL wing-body configuration (CRM-HL-WB), and was done in coordination with the AIAA High-Fidelity CFD Verification Workshop [20]. While the full CRM-HL configuration features surfaces typical of a commercial aircraft in landing configuration, including slats, flaps, nacelle, horizontal tail, vertical tail, and landing gear, the CRM-HL-WB configuration chosen for Test Case 1 removes most of this complexity, leaving just the wing and body. This configuration is shown in Fig. 1, and the test case parameters are given in Table 1.

**Fig. 1 CRM-HL-WB Configuration for Test Case 1**

The high level objective for test case 1 is to demonstrate grid convergence on the simplest configuration, before introducing the geometric complexities associated with high-lift devices. Beyond grid convergence, exploration of sensitivities to solver settings specific to particular methodologies (for example, turbulence model choice) is encouraged. It is the hope that lessons learned on this test case are directly applicable to the more complicated test cases that follow.

Key questions for test case 1 are:

- 1) Are RANS solvers able to demonstrate verification on this problem and series of grids?
- 2) For Non-RANS solvers, what is the most consistent approach to grid families that can demonstrate a trend towards grid independence on this problem?
- 3) Is there enough consistency among non-RANS approaches that there is reasonable agreement on a grid-independent solution?
- 4) Does the ensemble of results among modeling approaches compared to the experimental free-air corrected data tell us anything useful about uncertainty?

In order to meet the objectives of the High-Fidelity CFD Verification Workshop, the RANS participants were highly encouraged to run the SA-neg-R(crot1)-QCR2000 turbulence model[21], and utilize a consistent set of grids provided,

which are further described in Section IV.

For Non-RANS participants, particularly those utilizing scale-resolving techniques in the WMLES and HRLES TFGs, the standard approaches to verification become challenging due to the use of different grid topologies, temporal discretization techniques, duration of simulations and averaging windows, and even equation sets. As a result, there is typically not enough overlap in related results to meaningfully declare any one specific solution approach the standard. Non-RANS participants were encouraged to demonstrate grid convergence using multiple levels of grid, along with best practice solver settings, while monitoring integrated force and moment convergence. Gridding requirements for these TFGs were purposefully left vague, such that participants could uniquely define how to best design grid families to help answer the key questions.

Results from Test Case 1 were intended to be compared to full-model test data obtained using the Boeing 6% CRM-HL at the QinetiQ wind-tunnel facility. Due to unforeseen manufacturing delays, the test data was unavailable in time for the workshop. There was also no CRM-HL-WB test data available from other Ecosystem partners, including the ONERA F1 test heavily referenced for Test Case 2. Consequently, Test Case 1 was done entirely computationally, without a comparison to experimental data at this time. Future workshops may elect to revisit this problem after experimental data is collected.

B. Test Case 2: CRM-HL Configuration Buildup

Test Case 2 attempts to assess the ability of CFD to predict the effect of varying geometric fidelity through component build-up to help isolate specific types of flow physics associated with high-lift aerodynamics. The CRM-HL is designed such that components can be added to the wing-body configuration one at a time, to expose individual flow features systematically, in contrast to attempting to parse out all of the complex flow interactions from the full landing configuration. This test case additionally expands beyond test case 1 by requiring full lift polars consisting of 5 to 7 angles of attack, instead of just a single point.

Test case 2.1 begins with similar geometry to test case 1, the CRM-HL-WB configuration, but now includes the horizontal tail, vertical tail, and flap support fairings. Test case 2.2 adds a full span leading-edge slat to test case 2.1. This greatly increases the angle of attack at which the wing stalls, and exposes the wing to additional complex flow features such as a slat wake and pressure losses due to slat brackets. Test case 2.3 adds both an inboard and outboard trailing-edge flap with the same positioning as the landing reference geometry, serving to greatly increase the total circulation of the configuration. Finally, test case 2.4 adds the pylon and nacelle (with a chine), and is equivalent to the CRM-HL-LDG reference configuration. This series of configurations are shown in Fig. 2.

While Test Case 1 was intended to help develop modeling practices including grid refinement techniques and numerical modeling choices, Test Case 2 was chosen to help refine those practices. By directly comparing numerical solutions to experimental data in this systematic manner, one can begin to link simulation accuracy to the presence of a single geometric feature, and update modeling techniques developed to improve simulation accuracy. Key questions for Test Case 2 revolve around this approach:

- 1) Does the consistency in integrated forces and moments from CFD simulations improve when modeling geometrically simpler high-lift configurations?
- 2) If accuracy falls off due to the presence of a single component, can better modeling approaches be established to improve the predictions?

This test case is designed to utilize CRM-HL configuration geometry that matches experimental data collected for the equivalent as-designed geometry tested in the ONERA F1 wind-tunnel [22]. As mentioned previously, this experiment did not collect data on the wing-body configuration. Rather, it was intended that CFD results for Test Case 2.1 would be compared to data obtained from a test on the Boeing 6% model in the QinetiQ 5-metre tunnel. CFD simulations were requested to match the free-air data available from ONERA's wind-tunnel tests, with test parameters documented in Table 2. Further, Test Case 2 was conducted semi-blind, where experimental data for selected anchor points for configurations 2.2 through 2.4 were provided in advance. Once most initial results were computed and submitted, full datasets from the experiment were released to participants.

C. Test Case 3: Reynolds Number Study

Test Case 3 is intended to provide blind CFD predictions of wind-tunnel data collected across the CRM-HL ecosystem, particularly over the wide range of Reynolds number tested. Ecosystem data spans from the smaller scale atmospheric models operating at a chord Reynolds number of approximately 1 million, up to cryogenic models operating at a near-flight chord Reynolds number of 30 million. It was requested that participants explore mesh sensitivities, both

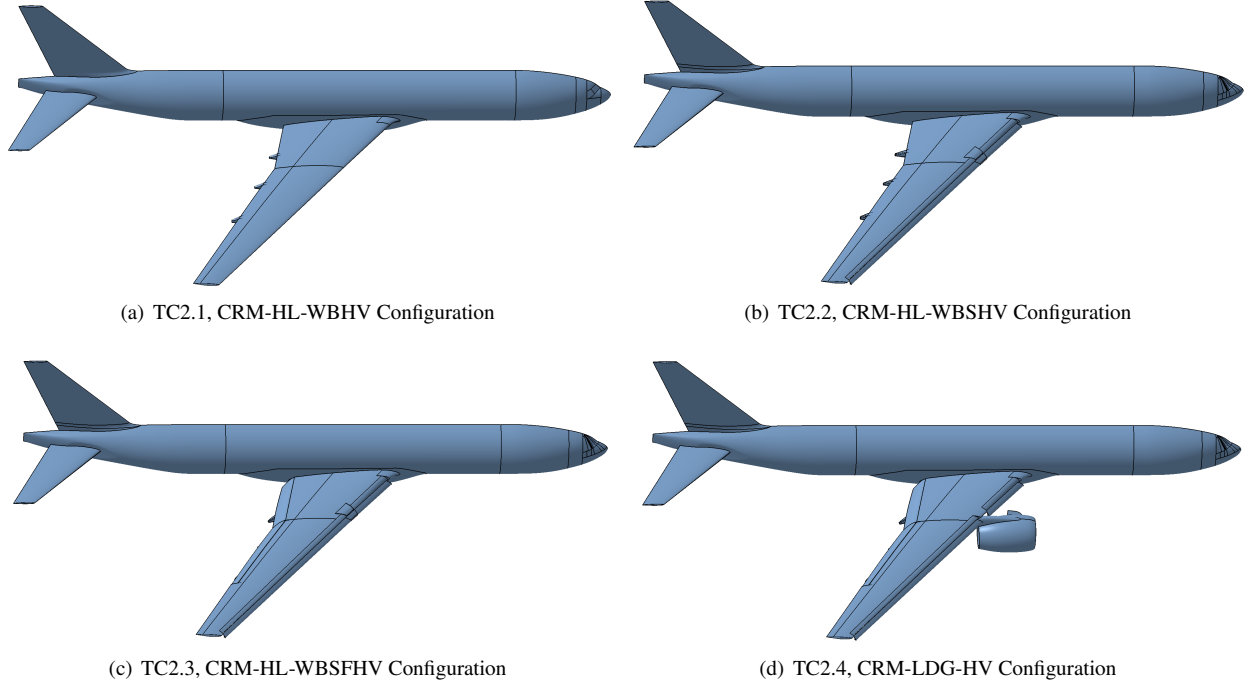


Fig. 2 Test Case 2 Geometric Configurations

near $C_{L,max}$ and in the linear C_L versus α range. Geometry for this test case was chosen to match the as-designed geometry of the NASA 5.2% model semi-span model, which was tested in the National Transonic Facility (NTF). It is the nominal CRM-HL-LDG configuration, very similar to what is shown in Figure 2(d), but updated to match the specific geometric details of the NASA 5.2% model, and with the horizontal and vertical tails removed. Though no single model can be used to collect data at the full Reynolds range required, a single nominal landing geometry was used for all Reynolds numbers simulated within the Ecosystem, with the expectation that model-to-model differences are small. The flow parameters for Test Case 3 are documented in Table 3.

This test case was chosen to further the understanding of modeling and gridding requirements as a function of Reynolds number. Rather than focusing on the accuracy of the CFD methods in predicting absolute integrated forces and moments, the objective is instead to predict increments with Reynolds number. Key questions for test case 3 are as follows:

- 1) How do gridding requirements change with Reynolds number?

Table 2 Test Case 2 Parameters

Domain	Hemisphere preferred
Domain Size	100x MAC
Mach	0.20
Chord Reynolds Number	5.4×10^6 (TC2.1) 5.9×10^6 (TC2.2 - TC2.4)
α	2.1: $6^\circ, 10^\circ, 12^\circ, 13^\circ, 14^\circ$ 2.2: $6^\circ, 10^\circ, 17.7^\circ, 20^\circ, 21.5^\circ, 23^\circ, 23.8^\circ$ 2.3: $6^\circ, 10^\circ, 14^\circ, 16^\circ, 17.7^\circ, 20.7^\circ, 23.5^\circ$ 2.4: $7.6^\circ, 10^\circ, 14^\circ, 16^\circ, 17.7^\circ, 19.7^\circ, 23.6^\circ$
Reference Static Temperature	518.67 °R

- 2) How does grid convergence behavior change with Reynolds number?
- 3) How consistent are CFD predictions with each other across a range of Reynolds numbers?

Table 3 Test Case 3 Parameters

Domain	Hemisphere preferred
Domain Size	100x MAC
Mach	0.20
Chord Reynolds Number	TC 3.1: 1.05×10^6
	TC 3.2: 5.49×10^6
	TC 3.3: 16×10^6
	TC 3.4: 30×10^6
α	6°, 10°, 14°, 16°, 18°, 19°, 20°, 22°
Reference Static Temperature	518.67 °R

Test Case 3 was intended to cover a range of experiments, but lean heavily on data from the NASA 5.2% model as tested in NTF, which was originally scheduled to occur prior to starting CFD studies associated with Test Case 3. Unfortunately, the wind-tunnel test was not conducted until late in the workshop, and the resulting experimental data did not correlate well with other Ecosystem data. As a result, it was decided to not directly compare the CFD results against experimental data, and instead conduct Test Case 3 blind.

IV. Overview of CFD Meshes

Over 245 individual meshes, comprising about 37 mesh families, were created and shared for this workshop. Most of these were made by mesh generation experts who were connected with one or more of the TFGs. Some meshes were also shared by participants. Mesh-generation guidelines from previous workshops were available for creating RANS meshes, but the mesh-generation experts were not constrained to follow these. In many cases, lessons learned during the TFG activities and virtual meetings led to new meshing approaches. All meshes shared by participants are available on the HLPW website [23], though several were used by participants that weren't shared, particularly those that didn't follow the traditional unstructured paradigms.

Figure 3 shows the mesh size ranges for the various families of grids made available for the three test cases in HLPW-5. For example, among the RANS grids created for Case 1 (red squares in the left-most section of the figures), the coarsest grids (open symbols and dashed lines) had around 1 million cells and 1 million nodes, while the finest grids (filled symbols and solid lines) had around 2 billion cells and over 300 million nodes. The meshes for Cases 2 and 3 tended to be somewhat larger than those for Case 1, due to the increased complexity of the configurations. The HO meshes were generally more coarse than the standard meshes created for the other TFGs; this is because HO meshes contain additional degrees of freedom within each of its cells. It is noteworthy that there were many meshes created with over a billion nodes or cells. The largest contributed mesh was a WMLES grid created for one of the Case 2 configurations, containing 6.2 billion cells and 2.3 billion nodes.

Most CFD practitioners are very familiar with RANS grids and gridding practices. RANS meshes are generally body-fitted, and tend to have very fine normal spacing at walls, typically with the goal of having a y_+ near unity at the first grid point or cell center (this spacing relaxes as you move away from the wall). The streamwise and spanwise spacings near walls are generally much larger, albeit with clustering in important regions like leading and trailing-edges. The end result is a mesh with cells near walls that are like pancakes, very flat in shape, with very large aspect ratios. Meshes for HRLES still require RANS-like fine spacing near walls, but in the regions where LES is expected to be active, the cells optimally should be fine enough to resolve the turbulent eddies and are more isotropic in shape. In contrast, WMLES modeling best practices suggest cells which tend to be fairly isotropic in shape (with aspect ratios typically less than 5) everywhere within the boundary layer or within active turbulent shear layers/wakes. A typical WMLES grid may have on the order of 20 cells across the boundary layer, although this can be very challenging to achieve (nearly impossible with today's resources) near the front of wings where the boundary layers are extremely thin.

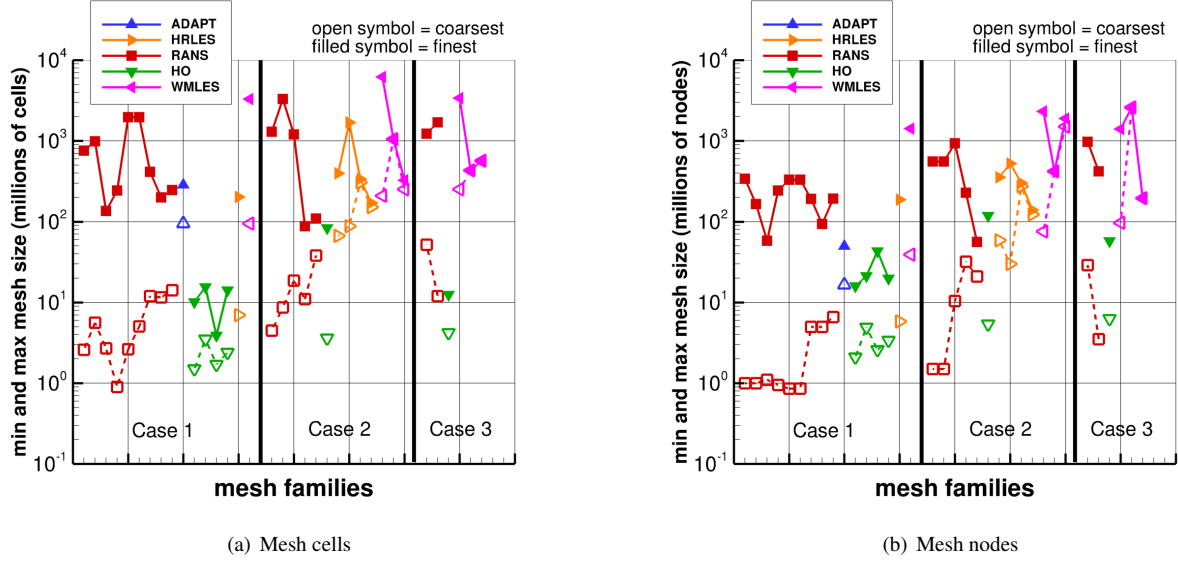


Fig. 3 Summary of HLPW-5 mesh size ranges

V. Summary of Entries

Over the course of the high-lift prediction workshops, the number of active participants and submitted test cases has steadily increased. In HLPW-1, 21 groups submitted 39 entries; in HLPW-2, 26 groups submitted 48 entries; in HLPW3, 35 groups submitted 79 entries; and in HLPW-4, 44 groups submitted 184 entries.

In HLPW-5, its three test cases were computed sequentially by the TFGs over more than a year leading up to the workshop. Some teams computed all three cases, while others did not. Two “mini-workshops” were held virtually to discuss interim results for Cases 1 and 2. Altogether, 47 groups submitted 365 entries, with 41 groups contributing to Case 1, 40 to Case 2, and 18 to Case 3. The entries are summarized in Table 4, where the particular TFG is denoted in the entry number by a letter: R for RANS, A for ADAPT, H for HO, L for HRLES, and W for WMLES. It should be noted that the WMLES TFG included two groups that did not use WMLES: W-001 used the unsteady Euler equations on adapted meshes and W-009 used a Lattice-Boltzmann scheme with a very large-eddy simulation (VLES) methodology.

Table 4 Summary of HLPW-5 Entries.

Entry number	Primary organization	Code name	Total number of entries	Test cases:								
				1	2.1	2.2	2.3	2.4	3.1	3.2	3.3	3.4
R-001	NASA LaRC	FUN3D	10	Y	Y	Y	Y	Y	-	Y	-	Y
R-002	Boeing	FELight	13	Y	Y	Y	Y	Y	Y	Y	Y	Y
R-003	NASA LaRC	USM3D-ME	12	Y	Y	Y	Y	Y	-	Y	Y	Y
R-004	DeMontfort U	OpenFOAM and Fluent	7	Y	Y	Y	Y	Y	-	-	-	-
R-005	Politecnico di Milano	SU2	10	Y	Y	Y	-	-	-	-	-	-
R-006	Siemens	STAR-CCM+	13	Y	Y	Y	Y	Y	-	-	-	-
R-007	Indian Inst. of Science	HiFUN	3	Y	Y	Y	-	-	-	-	-	-
R-008	Bombardier	DRAGON	18	Y	Y	Y	Y	Y	Y	Y	Y	Y
R-009	Kawasaki Heavy Indust.	Cflow	9	Y	Y	Y	Y	Y	-	Y	Y	Y
R-010	Aerodynamic Solutions	Leo	6	Y	Y	Y	-	Y	-	-	-	-
R-011	DLR	CODA	8	Y	Y	Y	Y	Y	-	-	-	-
R-013	Textron Aviation	FUN3D	6	Y	Y	Y	-	-	-	-	-	-
R-014	Zenotech	zCFD	2	Y	-	-	-	-	-	-	-	-
R-015	JAXA	TAS	17	Y	Y	Y	Y	Y	Y	Y	Y	Y
R-016	Embraer	CFD++	26	Y	Y	Y	Y	Y	Y	Y	Y	Y
R-017	Ansys	Fluent	10	Y	Y	Y	Y	Y	-	-	-	-
R-019	Synapsis-AG	Synapsis	1	Y	-	-	-	-	-	-	-	-
R-020	JAXA	FaSTAR	11	Y	Y	Y	Y	Y	-	-	-	-
R-021	Polytechnic U Montreal	CHAMPS	7	Y	Y	Y	Y	-	-	Y	Y	Y
R-022	Flexcompute	Flow360	9	Y	Y	Y	Y	Y	-	-	-	-
R-023	Luminary Cloud	Luminary	10	Y	Y	Y	-	Y	-	Y	Y	Y
R-024	ONERA	elsA	4	Y	Y	-	Y	Y	-	-	-	-
R-025	Cadence	Fidelity	4	Y	-	Y	Y	Y	-	-	-	-
A-002	Istanbul Tech U	HEMLAB-pyAMG	2	Y	-	Y	-	-	-	-	-	-
A-003	Boeing	EPIC-T1 and -BCFD	6	Y	Y	Y	-	Y	-	-	-	-
A-004	Inria	WOLF	7	-	Y	Y	Y	Y	-	-	-	-
A-006	Ansys	Fluent	5	Y	-	Y	Y	Y	-	-	-	-
H-005	U Tennessee	COFFE	1	Y	-	-	-	-	-	-	-	-
L-001	Flexcompute	Flow360	10	Y	Y	Y	Y	Y	-	-	-	-
L-003	Boeing	FELight	1	Y	-	-	-	-	-	-	-	-
L-004	AWS	CFD++	18	Y	-	Y	Y	Y	-	-	-	-
L-005	Ansys	Ansys	2	Y	-	-	-	-	-	-	-	-
L-009	Gulfstream	FUN3D	1	-	-	-	-	Y	-	-	-	-
L-013	USAF	KCFD	4	-	Y	Y	Y	Y	-	-	-	-
L-015	Numeric Systems	Pacefish	2	Y	-	Y	-	-	-	-	-	-
W-001 ¹	KTH	AdaptiveEuler	9	Y	-	Y	Y	Y	Y	Y	Y	Y
W-003	Boeing	BCFD	4	Y	-	-	-	-	Y	Y	Y	-
W-004	Boeing	CHARLES	11	Y	Y	Y	Y	Y	-	Y	Y	Y
W-005	NASA LaRC	FUN3D	12	Y	Y	Y	Y	Y	-	Y	Y	Y
W-006 ²	U Kansas	hpMusic	7	Y	-	Y	-	Y	-	Y	Y	Y
W-007	NASA Ames	LAVA	11	Y	Y	Y	Y	Y	-	Y	Y	Y
W-009 ³	Dassault Systemes	PowerFLOW	8	Y	Y	Y	Y	Y	-	Y	Y	Y
W-010	Volcano Platforms	ScaLES	20	Y	-	Y	Y	Y	-	Y	Y	Y
W-011	Tohoku U	FFVHC-ACE	3	-	-	-	-	-	-	Y	Y	Y
W-012	Scientific Simulations	NSU3D	3	Y	-	-	-	Y	-	-	-	-
W-013	Embraer	SU2	1	-	-	-	-	Y	-	-	-	-
W-014	Ansys	Ansys	1	-	-	-	-	Y	-	-	-	-

¹ Euler with adaptive meshing; ² high-order scheme; ³ Lattice-Boltzmann.

VI. Workshop Results

The workshop results are covered here in three sections, one for each test case. They are specifically addressed in terms of Key Questions (KQs) that were posed prior to the workshop’s activities. The KQs were written in order to focus on particular aspects of the computations, which, if answered satisfactorily, could significantly advance the high-lift community’s knowledge.

A. Test Case 1 Results

1. Are RANS solvers able to demonstrate verification on this problem and series of grids?

Case 1 does not yet have experimental data associated with it, but from the point of view of verification, this is not necessary, since we are only concerned with determining the “truth” solution for a given RANS turbulence model. The RANS TFG for this case ended up focusing primarily on results using the standard Spalart-Allmaras (SA), SA-neg and the SA-neg-R(crot1)-QCR2000 turbulence models (Note that the “neg” variant does not change the solution, so that the standard SA and SA-neg solutions are considered together). Other turbulence models were used by a few participants, but there were not enough data samples to be able to draw any conclusions about verification. However, for the two SA variants, a number of different codes demonstrated deep iterative convergence on all levels for a particular grid family, and subsequently showed grid convergence toward a common solution for each model.

Plots of all submitted RANS solutions (including different models and variants, solutions with poor iterative convergence, and solutions on non-optimal grid families) are shown in Fig. 4. The spread in the asymptoting solutions, as $N^{-1/3} \rightarrow 0$, was approximately $\Delta C_L = 0.02$, $\Delta C_D = 0.0040$, and $\Delta C_M = 0.02$. (A few outliers were even outside of this range.)

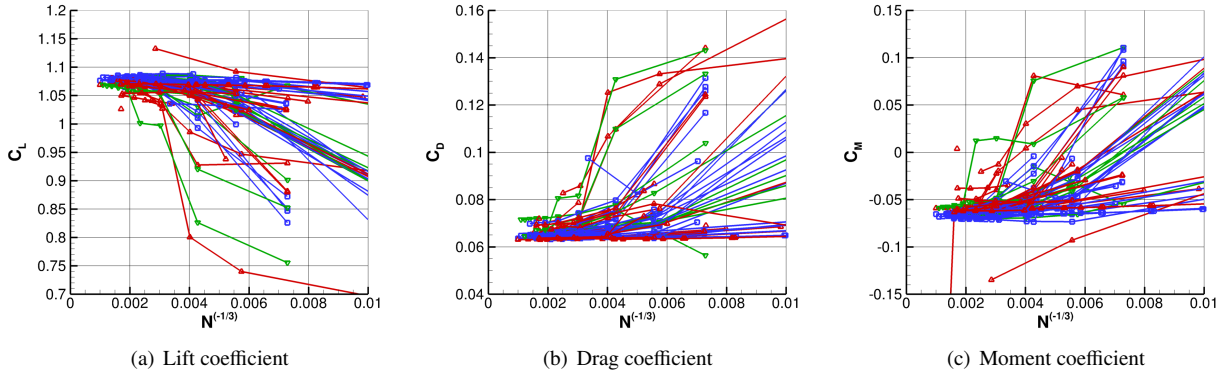


Fig. 4 All submitted RANS solutions for Test Case 1.

In contrast, codes that exhibited better mesh convergence trends tended to converge to a common solution. Results for the SA-neg and SA-neg-R(crot1)-QCR2000 models are shown in Fig. 5. For SA-neg, 12 different results were seen to asymptote toward a very narrow solution range for lift, drag, and moment as $N^{-1/3} \rightarrow 0$. The range was approximately $\Delta C_L = 0.005$, $\Delta C_D = 0.0005$, and $\Delta C_M = 0.003$. In the figures, different colored lines represent different mesh families, while RANS results on adaptive meshes are shown with a gradient symbol. These results indicate that different codes can display different grid convergence trajectories on the same mesh family. On the other hand, the trajectory can also be influenced by the particular mesh when using a given code. However, the particular trajectory is not important for verification; what matters is that the grid-converged answer is consistent for all codes as any given mesh is consistently refined.

For the SA-neg-R(crot1)-QCR2000 turbulence model, seven different participant results agreed well with each other as $N^{-1/3} \rightarrow 0$. The range was approximately $\Delta C_L = 0.004$, $\Delta C_D = 0.0006$, and $\Delta C_M = 0.002$. Notably, this model’s verified solutions were somewhat different from those of SA-neg, indicating that this verification process can distinguish between different turbulence model variants.

The answer to this KQ is yes, correctly coded RANS solvers are able to demonstrate verification on this problem, provided that the CFD results are sufficiently iteratively converged and run on appropriate, consistent grid families. However, it should be noted that we have only done verification checking on forces and moment. A limited check of

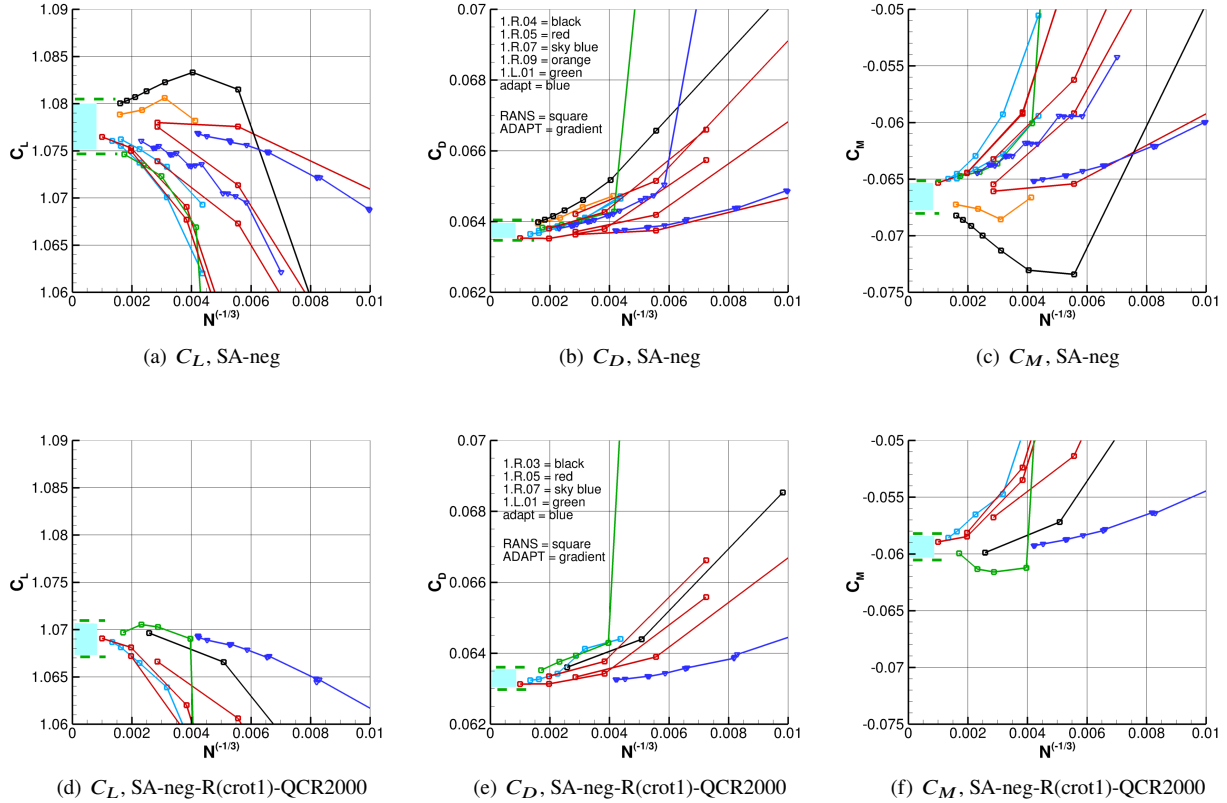


Fig. 5 Test Case 1 verified RANS results for SA-based turbulence models.

velocity and eddy viscosity profiles has revealed some areas, particularly near the wing trailing-edge and tip, with poor agreement between participant solutions. Unfortunately, many participants did not submit profile data, so it has not been possible to fully correlate good/poor agreement of forces and moment with good/poor agreement of profiles.

It is also worth noting that this test case is mostly attached flow, with only a very small region of separation near the upper surface trailing-edge. For a different test case with large region(s) of separation, it is likely that RANS results would have had more difficulty passing a verification test (because RANS consistency tends to deteriorate for separated flows). It is currently unclear at what angle of attack significant separation and stall would occur. Although not shown here, some RANS results indicate stall just beyond the current $\alpha = 11^\circ$, whereas others indicate much higher stall angle near 14 or 15° . Nonetheless, this is an important achievement to attain verified high-lift RANS solutions on a pre-stall 3-D wing-body configuration. With verified solutions in hand for two turbulence models, these established “truth” solutions can now be used by others to help verify their implementations of these models.

2. For Non-RANS solvers, what is the most consistent approach to grid families that can demonstrate a trend towards grid independence on this problem?

This QK ended up to be a difficult question to answer. Both HRLES and WMLES struggled to attain grid independent solutions for Test Case 1. Fig. 6(a) shows grid convergence results from the HRLES TFG. For this exercise, most of the participants used the same grid family (labeled as 1.L.01(2) on the HLPW website [23]), which was created using consistent mesh-refinement practices. As $N^{-1/3} \rightarrow 0$, the force and moment coefficients in the HRLES simulations did not asymptote to a fixed value. In fact, the lift dropped rather precipitously for most of the results on the finer grids in each family. This drop has been attributed to a failure of the standard DDES shielding function [24], which should treat attached boundary layers in RANS mode. This problem is sometimes referred to as “grid-induced separation.” Improvements to the shielding function, such as with the Deck-Renard method [25] used in the L-004.4 entry (solid red line), have demonstrated better grid convergence properties. However, there has not yet been a concerted effort among a

larger group of participants to determine if multiple codes can use a common improved shielding function to achieve consistent grid-independent DDES results for this case.

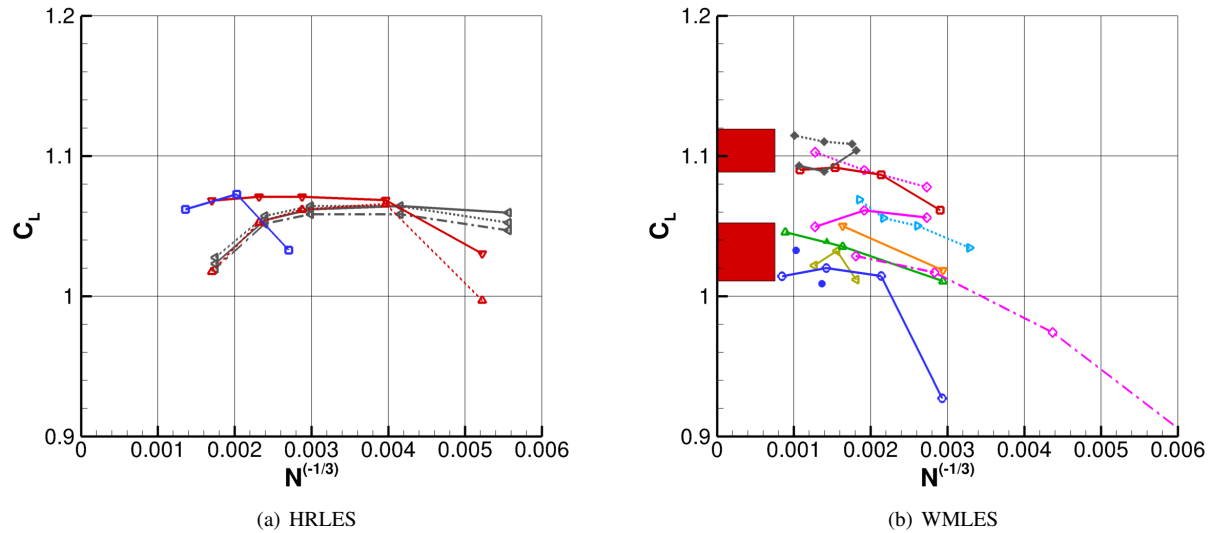


Fig. 6 Non-RANS grid convergence for Test Case 1.

The WMLES TFG also did not achieve grid-independent results for Test Case 1, as shown in Fig. 6(b). Although several individual entries showed some trend toward grid independence as $N^{-1/3} \rightarrow 0$, others did not. Also, there appeared to be trends toward two different solutions, indicated on the left side of the figure in red: high-lift near $1.09 < C_L < 1.12$ and low lift near $1.01 < C_L < 1.05$. Unlike the HRLES TFG, the participants of the WMLES TFG did not use any common grids or grid types. Each participant developed their own grid families, which included unstructured, Cartesian, and Voronoi types. No one type appeared to be more consistent than any other, and the reason for the solution bifurcation into a higher and lower lift branch is not known at this time.

Overall, then, the answer to this KQ is not known. Additional work is needed; the best practices for HRLES and WMLES are still evolving.

3. Is there enough consistency among non-RANS approaches that there is reasonable agreement on a grid-independent solution?

The answer to this question at this time is clearly no. Fig. 7 shows grid convergence trends for all TFG results, with the non-RANS results in green (HRLES) and pink (WMLES). Ignoring outliers, the red (RANS) results converge to a fairly tight band near $1.06 < C_L < 1.08$. The spreads in the HRLES and WMLES results are much larger, and their grid-converged results more uncertain, as discussed in the answer to the last KQ.

Additionally, we see larger variations in predicted surface pressure coefficients in the HRLES and WMLES results compared to RANS, as shown in Fig. 8 for row I, far outboard on the wing. In the two left plots, 53 independent RANS solutions show very little C_p variation for either the leading or trailing-edge regions. In Fig. 8(b), aside from two outliers, all HRLES solutions (which presumably are in RANS mode near the leading-edge) show similar agreement. But near the trailing-edge (Fig. 8(e)), the HRLES solutions show more variation. In the two right plots, the WMLES solutions show very large variations near both the leading and trailing-edges. It was noted during WMLES TFG discussions that the computed location of turbulence initiation on the wing could vary significantly, and this location likely played a role in the extent of trailing-edge separation predicted.

4. Does the ensemble of answers among modeling approaches compared to the experimental free air corrected data tell us anything useful about uncertainty?

The wind-tunnel experiment corresponding to Test Case 1 has not yet taken place. As a result, the answer to this KQ is inconclusive. However, we do know that the various CFD methods show notable variation in the solution, as

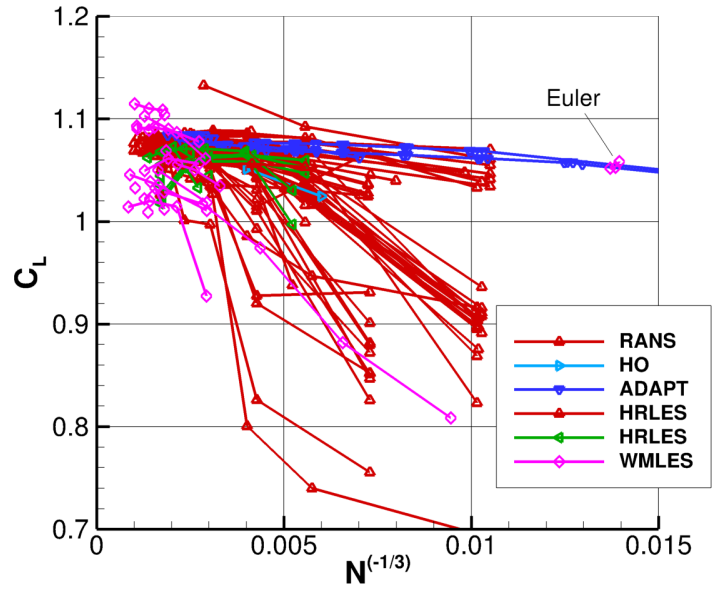


Fig. 7 Test Case 1 grid convergence trends for all TFG results.

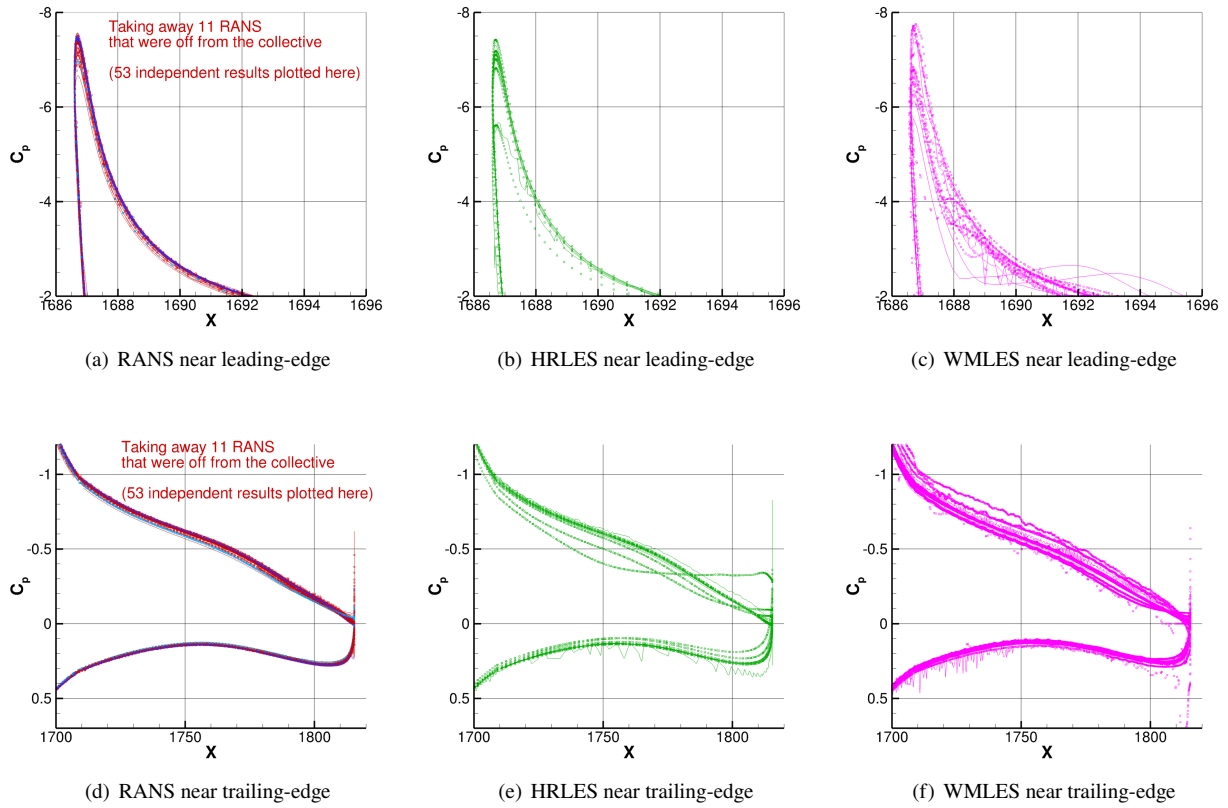


Fig. 8 Test Case 1 surface pressure coefficients, row I.

discussed above. The size and extent of trailing-edge separation also varies, as shown in Fig. 9. The view shown is looking outboard along the wing trailing-edge, from a position very close to the wing surface. RANS results (top row) show a very thin region of separation that varies slightly in shape, depending on whether the SA-neg or SA-neg-R(crot1)-QCR2000 is used. The typical HRLES result (Fig. 9(d)) is fairly similar to the RANS results with R and QCR included (Fig. 9(b)). As discussed earlier, WMLES results yielded two distinct “branches,” one with lower lift and one with higher lift. In Fig. 9(e), an example of a lower-lift solution is shown, with larger trailing-edge separation. Fig. 9(f) shows a higher-lift solution, with very little trailing-edge separation.

Based solely on this computational exercise, we now know that measurements beyond surface pressures and integrated forces and moments would be helpful for evaluating the CFD capabilities. When the wind-tunnel experiment for this test case is conducted, the following measurements should also be taken: untripped transition location, sensitivity to trip dots, surface flow visualization, and sensitivity to angle of attack and Reynolds number.

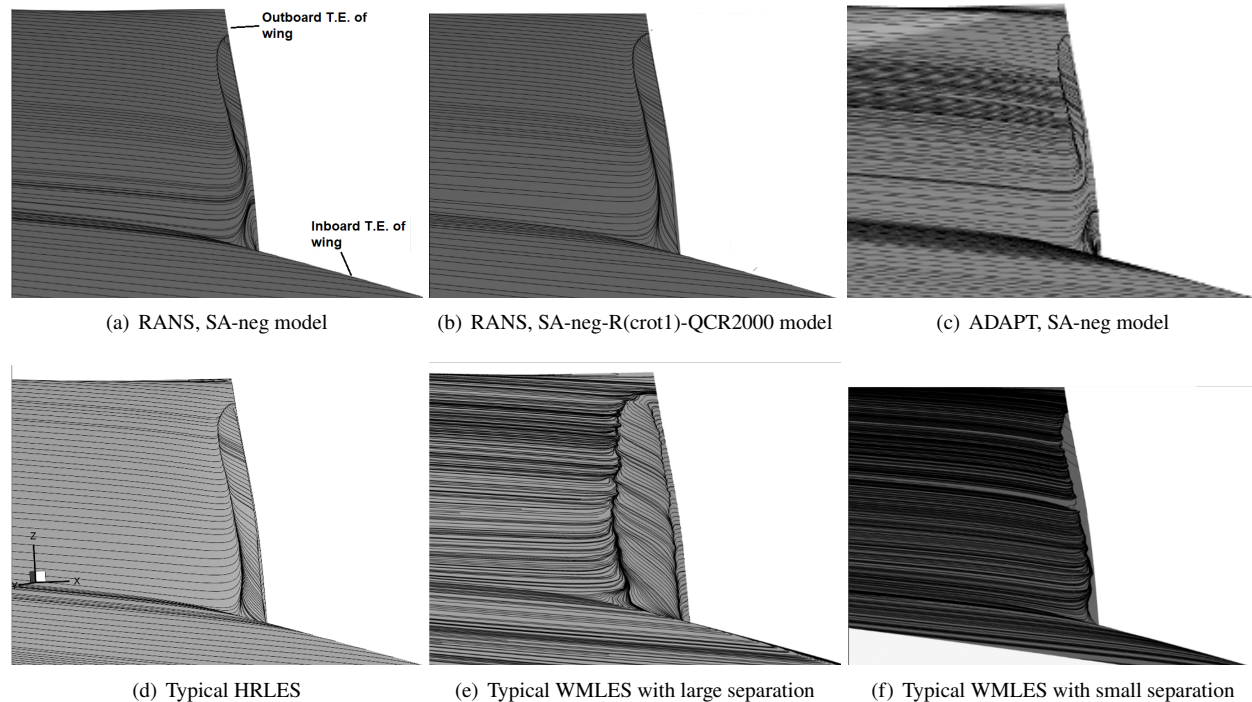


Fig. 9 Test Case 1 surface streamlines near the wing trailing-edge.

B. Test Case 2 Results

Test case 2 was conceived to study the ability of CFD to predict the incremental aerodynamic effects of a high-lift component buildup. As stated above, there were four configurations provided to participants: a simple wing-body (case 2.1), the wing-body with the addition of the leading-edge slat (case 2.2), the wing-body-slat with the addition of the trailing-edge flap (case 2.3), and the nominal landing configuration with the high-lift devices and the nacelle/pylon installed (case 2.4). Because there is experimental data available for test cases 2.2, 2.3, and 2.4, fewer participants performed CFD studies for test case 2.1. Also, due to a fewer number of submissions, it was generally more difficult to judge the results from the mesh-adaptation CFD simulations relative to the other methods.

1. Does the consistency in integrated forces/moments from CFD simulations improve when modeling geometrically simpler high-lift configurations?

One of the main objectives of this test case was to determine if the consistency of CFD results was better for the simpler configurations, based on the general expectation that CFD would be able to better model less complex flow physics. From the results presented at the workshop, the answer to this question largely depended on the specific

configuration considered. For test case 2.1, most RANS simulations predicted similar results at angles-of-attack below 13 degrees (see Fig. 10). However, at higher angles-of-attack, the CFD results diverged, with several simulations indicating varying amounts of flow separation at the wing trailing-edge near the wing planform break (see Fig. 11). At maximum lift and beyond, there was no consistency at all, which is not surprising due to the large amount of smooth body separation at this flow condition. For WMLES methods, the effects of boundary layer transition and mesh resolution were determined to be important, and likely drove significant differences seen in the various submissions (see Fig. 12). For the workshop, this test case unfortunately did not have any accompanying test data, but test data is expected to be acquired in future CRM-HL testing campaigns.

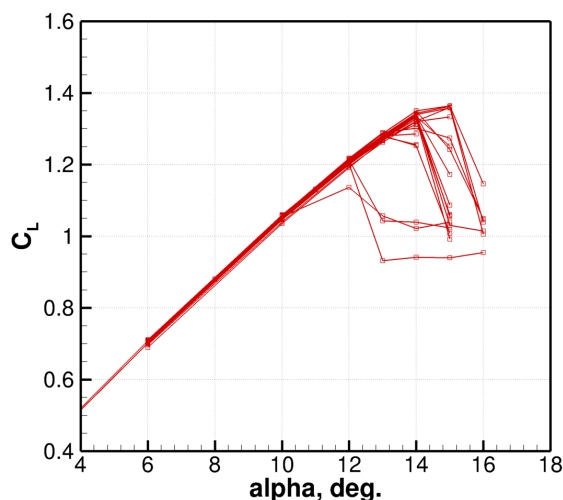


Fig. 10 RANS results for Test Case 2.1.

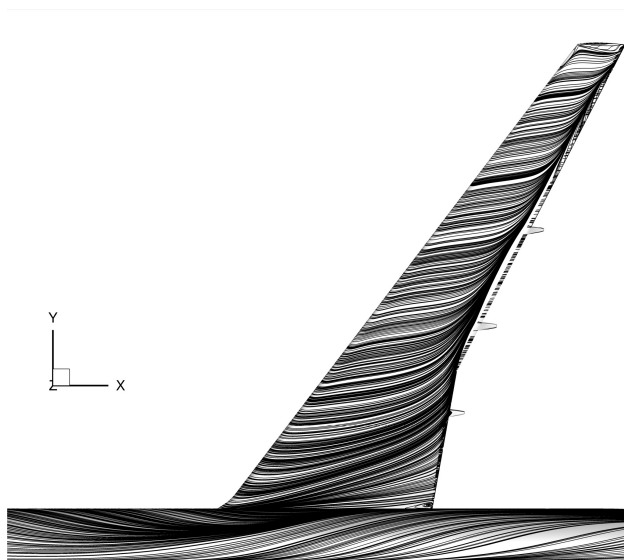


Fig. 11 Surface streamlines from RANS submission R-003 utilizing mesh 2.R.01.R.

For test case 2.2, the addition of the leading-edge slat component generally improved the consistency of simulation results, particularly when compared with available experimental data. Plots of scatter in C_L in the CFD results for all methods compared with test data at angles-of-attack of 10° and 23° are shown in Figure 13. For the RANS simulations, good consistency is obtained at $\alpha = 10^\circ$, but not near stall, at $\alpha = 23^\circ$. Based on a limited set of simulations, the use

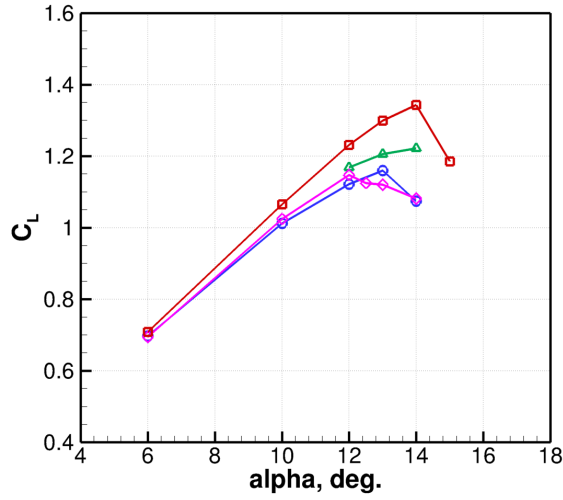


Fig. 12 WMLES results for Test Case 2.1.

of mesh adaptation, improved consistency for the RANS simulations at both the lower and higher angles-of-attack. HRLES and WMLES methods showed relatively good consistency at both angles-of-attack. These methods better predicted the lift magnitude near stall, as expected, but predicted too much lift compared to the RANS simulations at $\alpha = 10^\circ$. At $\alpha = 23^\circ$, both HRLES and WMLES methods showed an average of approximately 0.15 scatter in C_L between submissions compared to the approximately 0.5 scatter in C_L observed in the RANS datasets.

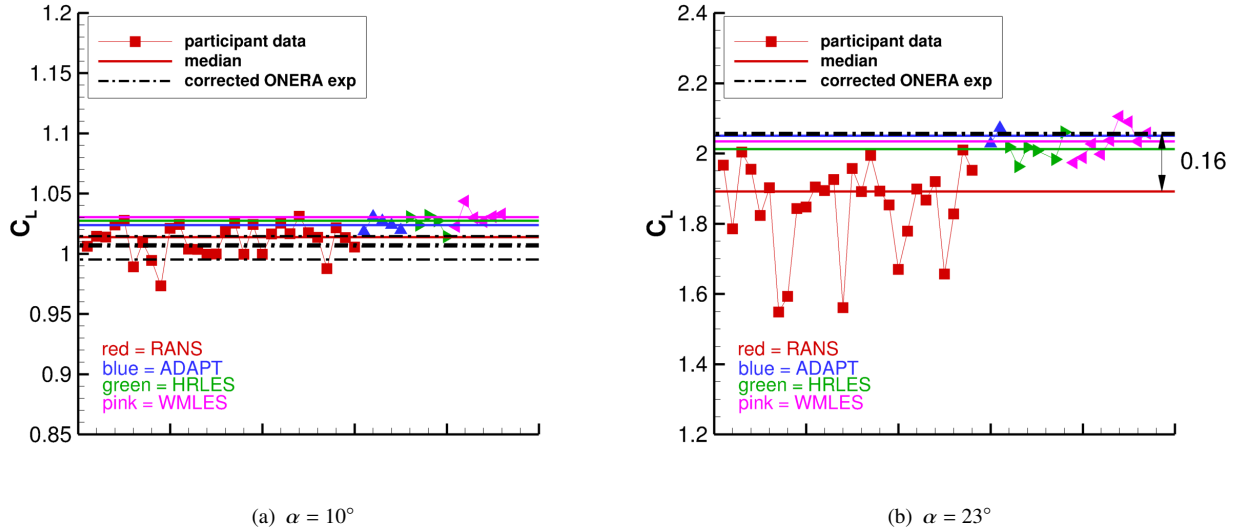


Fig. 13 Scatter in C_L for Test Case 2.2, all methods.

For test case 2.3, the differences in CFD predictions and test data were generally the highest among the four configurations studied for test case 2. Across all methods, inclusion of the flap negatively affected predictive accuracy, as well as reduced consistency between the CFD simulations at all angles-of-attack. For RANS simulations, there was a greater amount of scatter in the predicted lift values for test case 2.3 relative to test case 2.2 as shown in the lift curve comparison in Fig. 14. (Note that for test case 2.3, two of the RANS results performed reasonably well in terms of lift near $C_{L,max}$, but these methods, which used the SA model, both used their own grids and did not correlate well with the

verified SA results for test case 1. So the reason for their improved result on this test case is not known. These may be examples of “right answer for wrong reasons,” which have been noted from RANS in past workshops.) For WMLES simulations, a similar trend of more scatter for test case 2.3 compared to test case 2.2 generally holds true, as shown in Fig. 15.

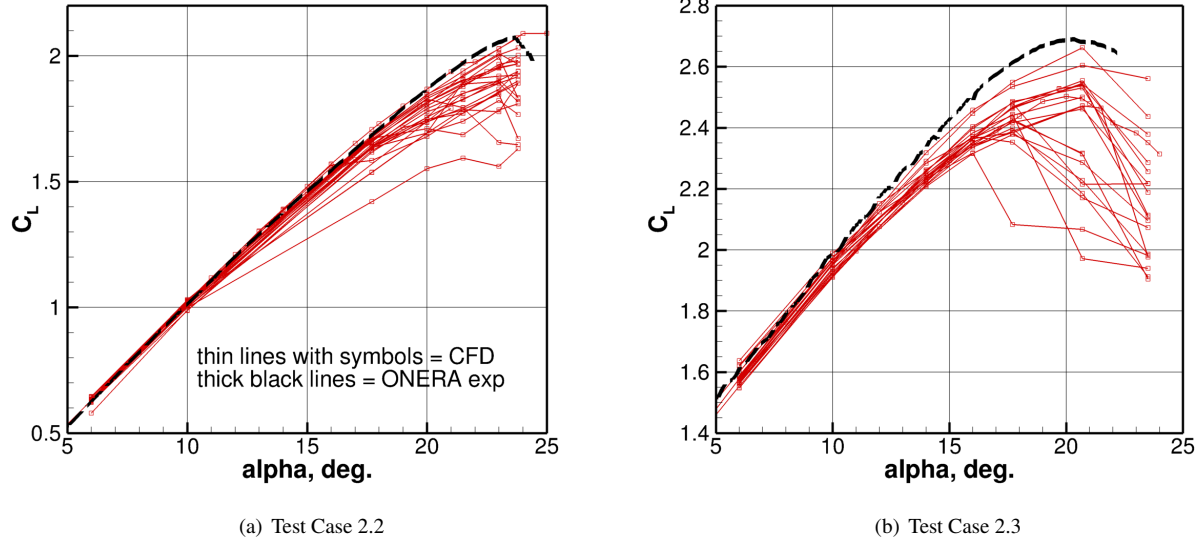


Fig. 14 Selected RANS results for Test Case 2

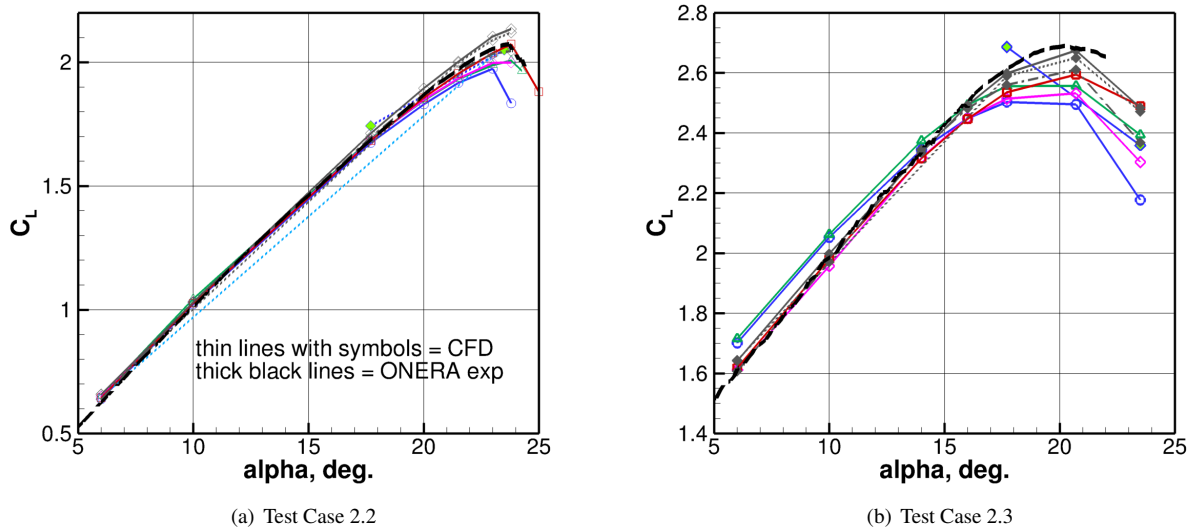


Fig. 15 Selected WMLES results for Test Case 2

For the nominal landing configuration in test case 2.4, which includes the addition of the nacelle, chine, and pylon, consistency in the CFD results did not appreciably improve for any of the methods utilized, particularly near maximum lift. As with the previous build-up configurations, the scale-resolving methods (HRLES and WMLES) better predict the lift magnitude at $C_{L,max}$ relative to experimental data (see Fig. 16).

The answer to this KQ currently depends on the CFD method used. For RANS, the consistency is somewhat better for the simpler configurations in Test Case 2.1 and 2.2. For WMLES, the consistency was best for Test Case 2.2. It

degraded on more complex configurations, but was also poor for the simplest wing-body configuration. There were not enough HRLES submissions to evaluate its consistency across configurations.

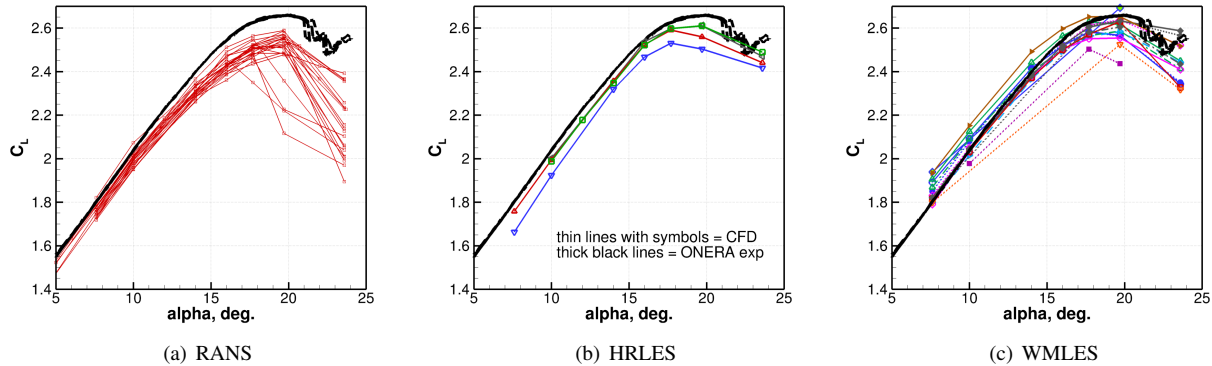


Fig. 16 Selected CFD results for Test Case 2.4.

2. *If accuracy falls off due to the presence of a single component, can better modeling approaches be established to improve the predictions?*

The answer to this KQ also depends on the CFD method used. For RANS methods, there was no convincing evidence that RANS turbulence models can reliably predict separated flows. However in studying the “pizza slice” wedge separation patterns, evidence suggests modeling approaches for this type of separation may be heavily influenced by the level of residual convergence in the simulation. For HRLES, simpler test cases exposed known weaknesses in mesh convergence using current DDES shielding function methods, particularly for flows where there are strong adverse pressure gradients. The Deck-Renard function was utilized by several participants and was shown to improve the robustness of the DDES method for Test Case 1 as the mesh was refined (see solid red line in Fig. 6(a)).

For WMLES, the systematic underprediction in CL_{max} from many participants led to increased attention to boundary layer transition initiation and modeling, and particularly the importance in modeling the correct boundary layer on the slat near CL_{max} . Focus on this area led to the implementation and testing of transition sensor models in several codes. In general, the inclusion of a transition sensor in WMLES modeling dramatically impacts results near CL_{max} , as shown in Fig. 17 from Test Case 2.4. However, there is still much work to be done to determine whether these sensor approaches provide a general predictive capability.

3. *Are there any conclusions that can be made regarding accuracy of one particular method over another?*

The answer here is yes. One good example where there is a clear superior method over others is in the prediction of post-stall aerodynamics. When accurately predicting the flow post stall, it is critical that the simulation properly models not only the integrated forces and moments, but also the correct mechanism by which the flow breaks down due to flow separation. In this case, the scale resolving methods (either HRLES or WMLES) are clearly superior. All RANS simulations, including those employing mesh adaptation, failed to capture the correct flow features, while the HRLES and WMLES results properly model the inboard flow separation and slat bracket wake behavior associated with the flow breakdown mechanism observed in the experimental data (see Fig. 18).

C. Test Case 3 Results

1. *How do gridding requirements change with Reynolds number?*

This question is somewhat related to the next one. The only participants who submitted data for Test Case 3 were in the RANS TFG and the WMLES TFG, and there were far fewer participants than for Test Cases 1 and 2. As a consequence, there was little or no coordinated exploration of the effects of gridding strategies for different Reynolds numbers. In addition, without experimental data to compare against, this KQ and the next one are difficult to effectively evaluate. At this point, the answer remains unknown.

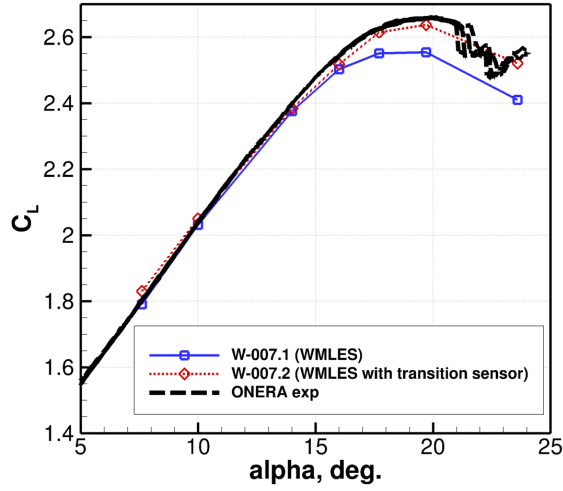


Fig. 17 Lift curve for WMLES for Test Case 2.4 showing impact of transition sensor function.

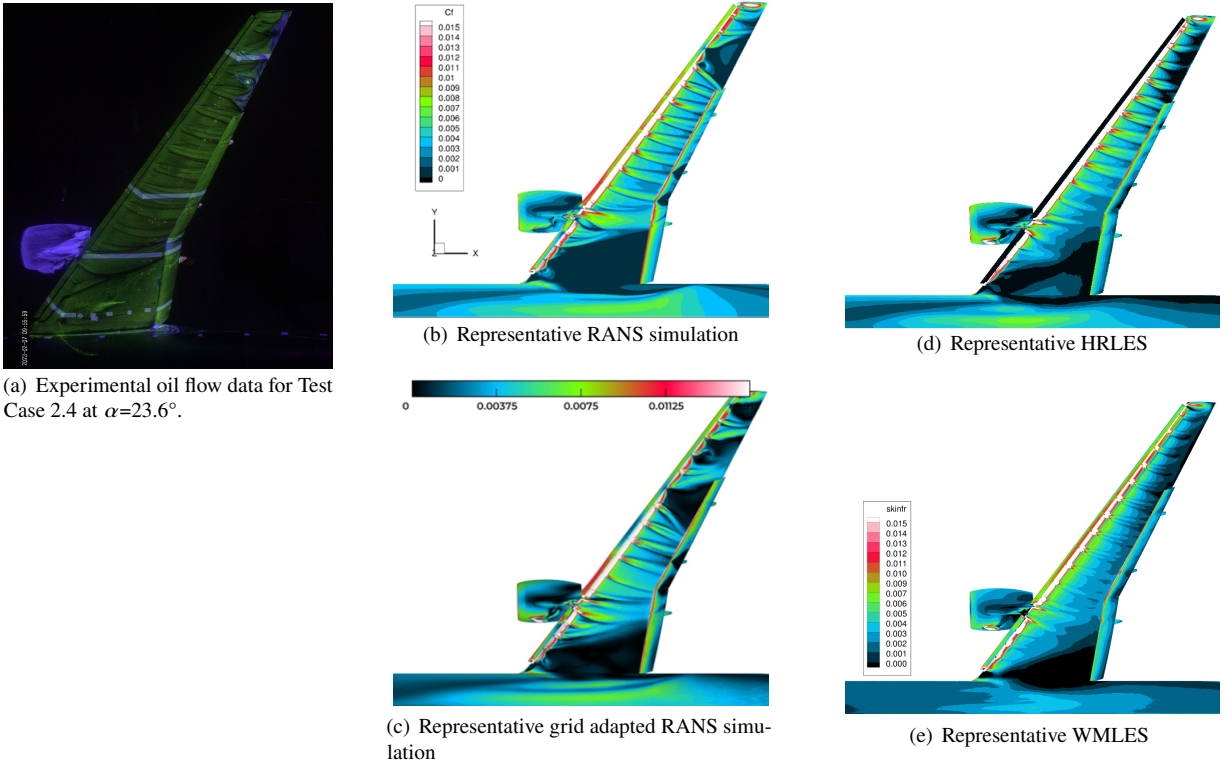


Fig. 18 Comparison of Experimental Oil Flow and selected CFD simulations for Test Case 2.4 at $\alpha=23.6^\circ$.

2. How does grid convergence behavior change with Reynolds number?

Fig. 19 shows grid convergence behavior of C_L for all submitted CFD results for all sub-cases at two different angles of attack: 6° and 18° . RANS results are in the top row and WMLES results in the bottom row. For RANS, simulations at neither angle of attack exhibited grid convergence (results are still changing notably even for the finest grids as $N^{-1/3} \rightarrow 0$). However, overall the results at $\alpha = 6^\circ$ appear to have a shallower slope than those at 18° . It has been noted

in the RANS TFG discussions that it was difficult to fully iteratively converge this test case. As a result, these grid convergence plots may actually mean very little.

For WMLES at $\alpha = 6^\circ$, there are only two sets of data. The lift for one set is begins to asymptote as the mesh is refined, but to very low levels. The other set of results appears relatively flat on the two coarser meshes, then decrease on the fine mesh, indicating non-monotonic behavior. On the other hand, some of the WMLES results at $\alpha = 18^\circ$ appear to be asymptoting, to some degree, as the mesh is refined. As a whole, RANS does not exhibit grid convergence at either angle of attack, while WMLES suggests that grid convergence is achievable at the higher angle of attack with currently available compute resources. Although without valid wind-tunnel results to compare against, it is hard to draw conclusions beyond grid convergence.

Interestingly, there are no distinctly different trends with Reynolds number in any of these results. Does this indicate that no additional (finer) mesh requirements are needed for higher Reynolds numbers? For WMLES, one might expect that finer mesh resolution would be needed to adequately resolve the finer turbulent scales and thinner boundary layers that occur at higher Reynolds numbers. The current data appear to be insufficient to make an informed assessment here. Therefore, the answer to this KQ is unclear.

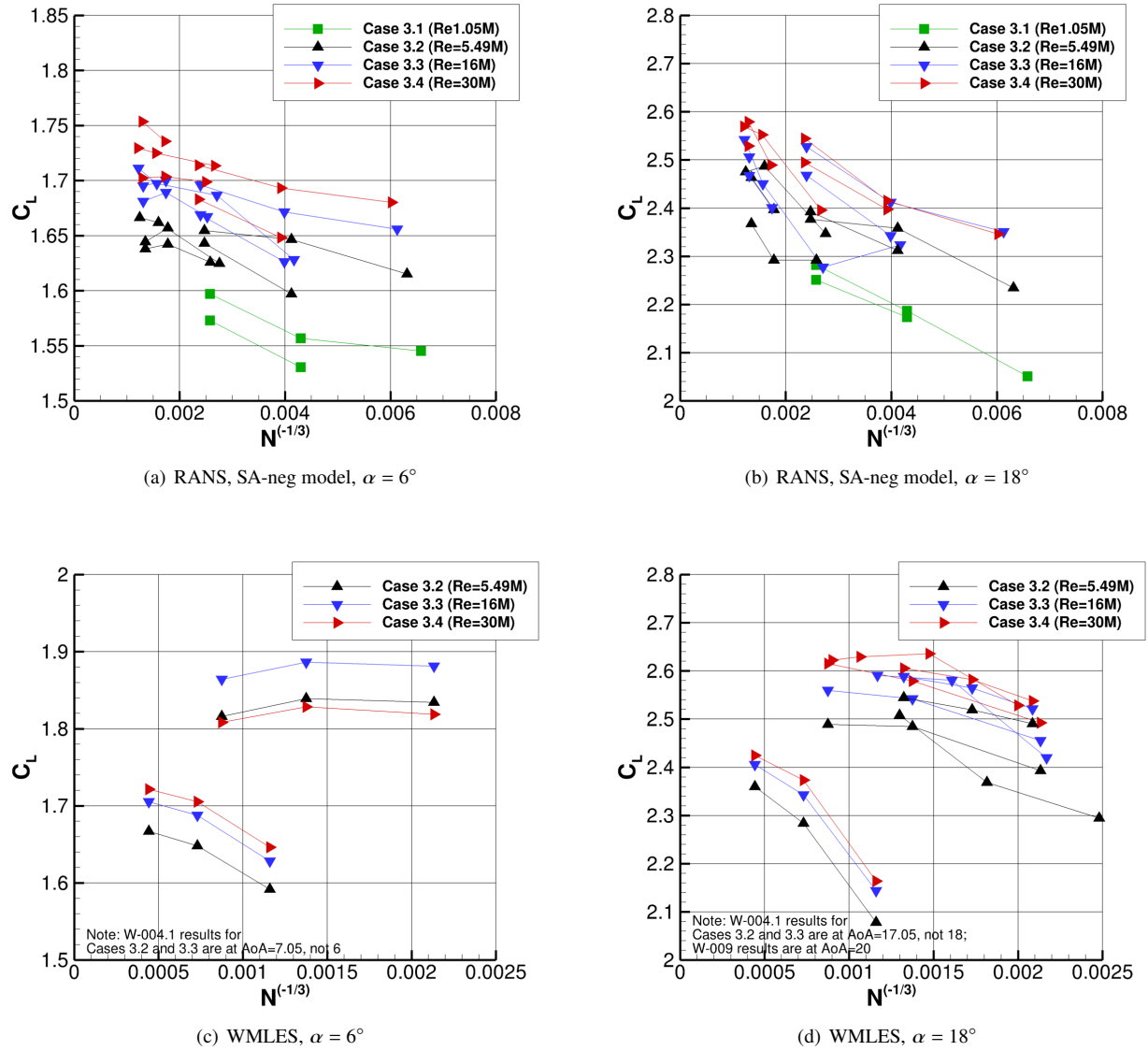


Fig. 19 Grid convergence of C_L for Test Case 3.

3. How consistent are CFD predictions with each other across a range of Reynolds numbers?

Comparisons of CFD force and moment polars for Test Case 3 are plotted for successively increasing Reynolds numbers in Figs. 20, 21, 22, and 23. Overall, the WMLES results appear to be somewhat more consistent (with each other) than RANS results near $C_{L,max}$, whereas the opposite is true at low angles of attack. Ignoring the lowest Reynolds number results in Fig. 20 (because there were only a few entries), the consistency of the solutions appear to be fairly similar across all Reynolds numbers. In other words, the Reynolds number has no discernible effect on the CFD inconsistencies exhibited.

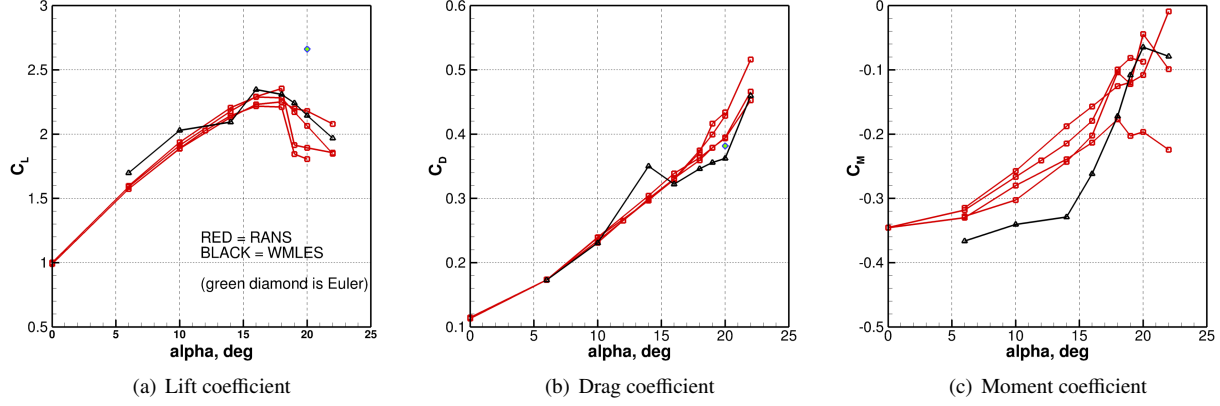


Fig. 20 Computed Test Case 3.1 polars, $Re = 1.05 \times 10^6$.

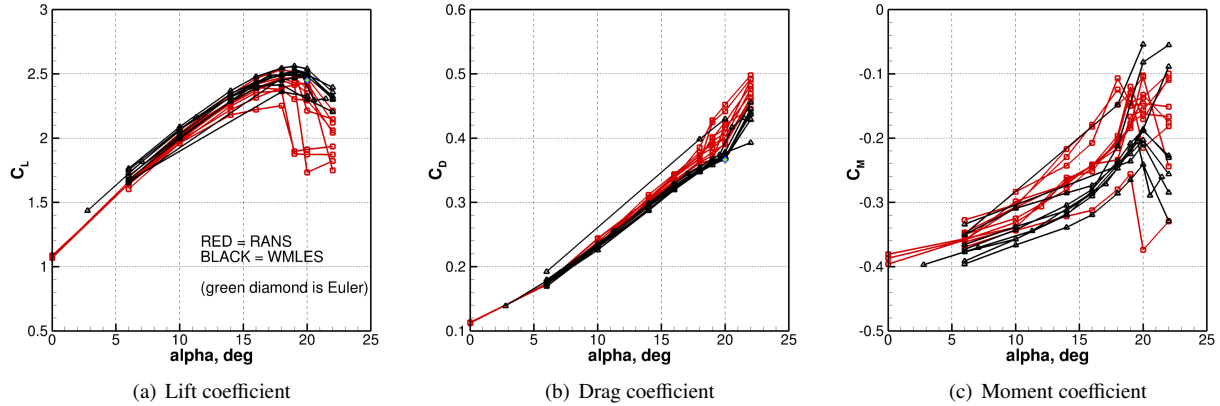


Fig. 21 Computed Test Case 3.2 polars, $Re = 5.49 \times 10^6$.

Fig. 24 shows the C_L values predicted at $\alpha = 10^\circ$ and 19° at the higher angle of attack near $C_{L,max}$, with outliers removed. Despite the inconsistencies in the absolute values predicted, the CFD results show a fairly consistent *trend* of increasing, asymptoting lift in the range $5.49 \times 10^6 < Re < 30 \times 10^6$. Although there is no experimental data available yet for this test case, this trend with Reynolds number is similar to what is expected based on other published results [26–29]. There is not enough consistency in the results at $Re = 1.05 \times 10^6$ at $\alpha = 19^\circ$ to discern a CFD trend at this Reynolds numbers.

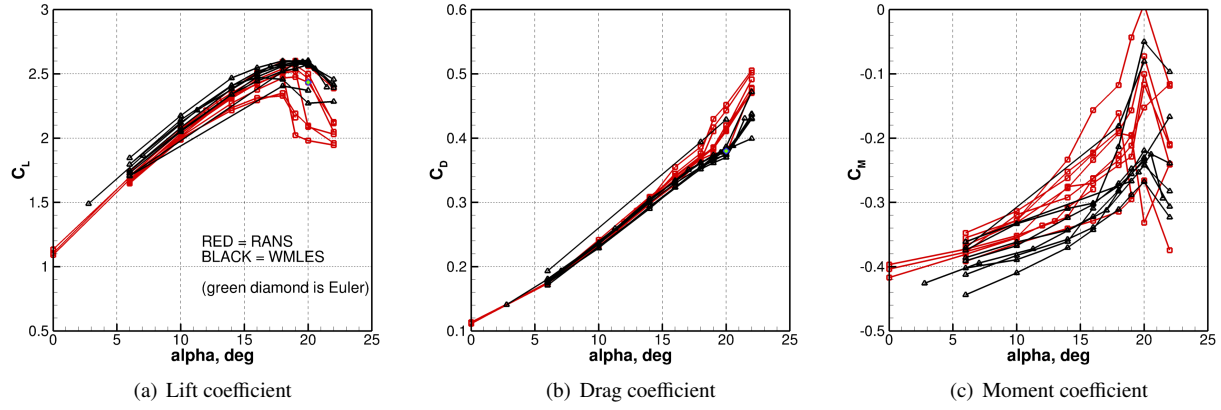


Fig. 22 Computed Test Case 3.3 polars, $Re = 16 \times 10^6$.

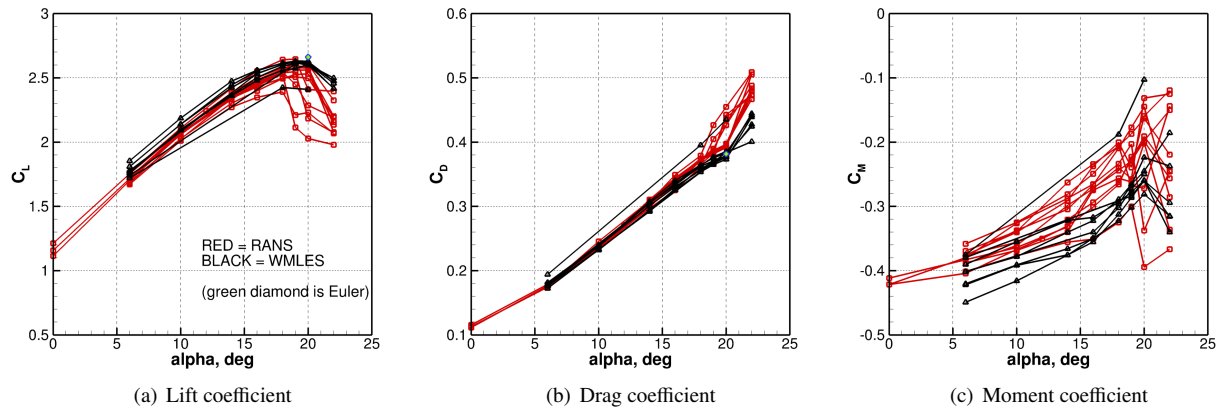


Fig. 23 Computed Test Case 3.4 polars, $Re = 30 \times 10^6$.

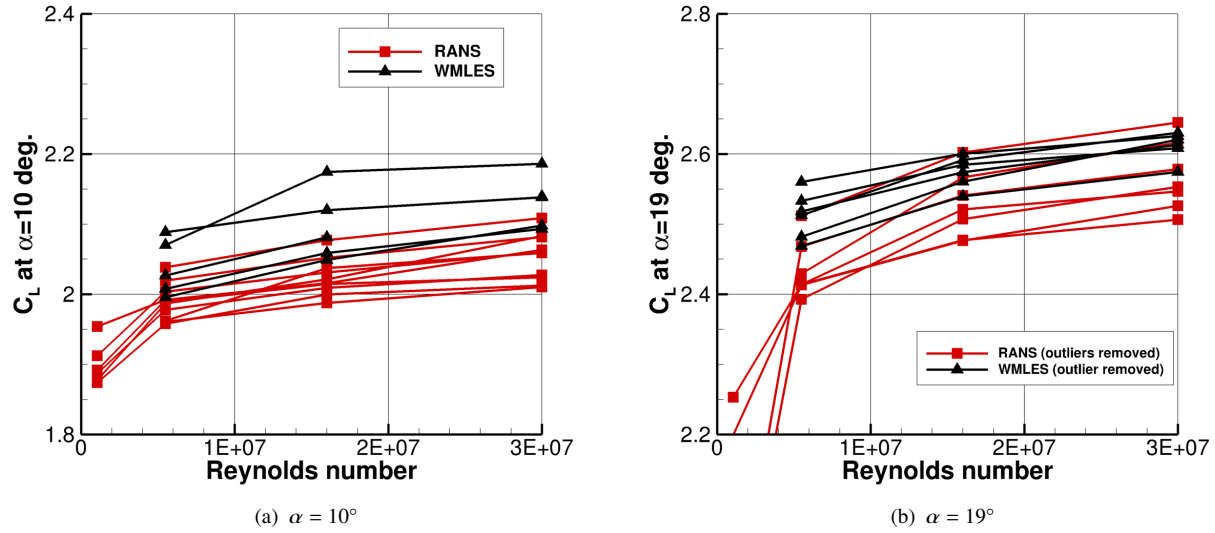


Fig. 24 Test Case 3 lift coefficient as a function of Reynolds number.

VII. Key Findings

The fifth High-Lift Prediction Workshop set out to answer a number of key questions discussed in Section VI. In taking a step back from the details, there are a number of key findings that are beyond the scope of the original key questions. This section highlights these additional findings by test case.

A. Test Case 1 Key Findings

In looking at trends over the series of five workshops since 2010, progress towards consistency using RANS methodologies is considerable. In the early High-Lift Prediction Workshops, participants had not yet converged on gridding settings, solver settings, or convergence criterion, nor had the importance of verification of turbulence model implementation been realized. As a result, spread in C_L , even in the linear region, could be upwards of 0.2, as shown in Fig. 25. In the present workshop, test case 1 was purposely chosen to be challenging, at the edge or beyond the linear range of the lift curve where the flow is on the verge of separation. Despite the added challenge, participants in the RANS TFG were able to demonstrate consistency in lift across solvers to within 0.005 in C_L . This is a remarkable improvement since the early workshops, and allows for robust conclusions to be made about modeling techniques and best practices without being clouded by uncertainties due to gridding, model implementation, and convergence.

Beyond RANS, this test case proved more challenging than initially intended. While it was originally selected to focus on grid convergence on a relatively simple problem, this test case exposed a number of modeling challenges for scale-resolving solvers. The scale-resolving solutions consistently identified trailing-edge flow separation but revealed significant sensitivity to modeling choices, which affected the extent of separation. As shown in Fig. 9, WMLES showed grid convergence towards two distinct solution families, each with varying degrees of trailing-edge separation. These variations, largely driven by differences in modeling choices that influence boundary layer development, are further explored in [14]. Unlike RANS, which has been extensively used in previous workshops, scale-resolving techniques have only been rigorously applied and evaluated since HLPW-4. As a result, there has not yet been sufficient time to develop and demonstrate consistent modeling approaches for this class of problems using these methods. Interestingly, a test case that consistently achieves a single solution across solvers in RANS can be much more challenging for scale-resolving methods, due to sensitivities to factors not typically modeled in RANS, such as boundary layer transition.

B. Test Case 2 Findings

From the outset [16, 17], the CRM-HL was designed to be modular, by facilitating the addition and removal of geometric components to systematically assess the impact of isolated flow features on vehicle aerodynamics. To date, test campaigns at ONERA [22], and NASA [30] have collected experimental data on these configurations. Test Case 2

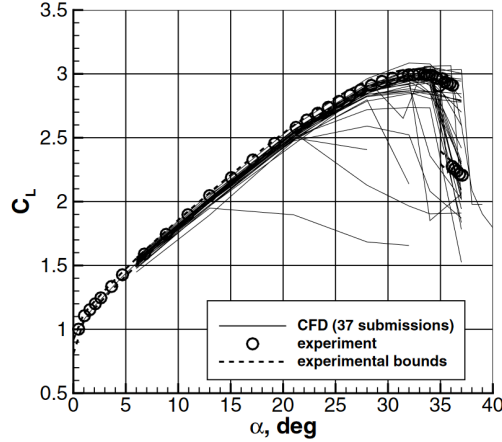


Fig. 25 Lift curve from HLPW-1, on the NASA Trapping configuration.

marks one of the first systematic approaches to study component build-up effects. Some of these results are highlighted in Section VI, but it is worth highlighting the general struggles of all methods when it came to predicting the effects of separated flow on the trailing-edge flap, as was shown in Fig. 14 and 15. This flow feature was previously studied through the flap deflection test case in HLPW-4 [4], and is also specifically targeted in the current Test Case 2. These cases conclusively demonstrate that there is much more work to be done to understand how best to capture the flow environment present on deployed trailing-edge flaps at moderate angles of attack.

Documented in section VI and emphasized here, is the importance of boundary layer transition for scale resolving methods. Participants in this workshop demonstrated that when using wall modeled approaches, the flow breakdown over the wing at $C_{L,max}$ is very sensitive to the development of the boundary layer over the leading-edge elements. Treating a portion of the slat boundary layer as laminar resulted in higher maximum lift, in better agreement with the experiment, as previously shown in Figure 17. This sensitivity highlights the strong need for experimental data focused on characterizing the laminar flow extent on the leading-edge slats. The evidence from WMLES participants, along with results from the early workshops [1, 2], suggests that a deeper focus on transitional behavior is required in the future, as it is likely that the sensitivity demonstrated in the WMLES TFG is not strongly associated with wall modeling, rather it is inherent in high-lift flow physics.

Lastly, it is worth mentioning the high computational cost of entry for this workshop. Across all test cases, workshop organizers requested solutions for 51 different test points across 6 unique geometries, along with sensitivity studies to both modeling parameters and gridding. This workshop demonstrated the value in computing the full lift polar as opposed to a subset of individual points, but it is also acknowledged that the number of test points was very large. Because of the availability of such a wealth of computational results, much more effort is required to adequately mine the data and fully digest what can be learned from these CFD studies. To this end, future workshops will need to reevaluate the quantity of test cases and/or data requested, and strongly consider revisiting test cases from previous workshops to allow for deeper focus.

Scale resolving simulations, in particular, have continued to show great promise in predicting aerodynamic performance across the range of test cases in this workshop, but simulations using these methods come at a high computational cost, often orders of magnitude above RANS methods. Grid convergence studies completed on some of the test cases in this workshop clearly demonstrate that there is little value in running under-resolved scale resolving simulations, as the results can trend away from the experimental data in a non-linear manner. It must be acknowledged that these approaches, while increasingly superior to RANS, are not yet affordable for all workshop participants.

C. Test Case 3 Key Findings

One of the objectives of Test Case 3 was to better understand CFD gridding requirements as a function of Reynolds number. For most computational methods that solve to the wall, and for which the first grid cell spacing is typically set to y_+ of 1, the question about grid resolution is fairly well understood. However, for approaches using wall modeling it is less so. There are many estimates in the literature for wall modeled gridding requirements [31], typically suggesting that the number of cells required will increase significantly as the Reynolds number increases. Although there was no

experimental data to compare against, a few participants in the WMLES TFG were able to demonstrate comparable trends in grid convergence between Reynolds numbers of 5.49 and 30 million. Simply put, grid convergence for simulations run at the two Reynolds numbers on the same grids appeared to be roughly equivalent (see Figure 19(c) and (d)). This finding is inconsistent with previous estimates, as well as physical intuition, and therefore warrants further attention in future workshops. Analysis of test case 3 computed boundary layer profiles may also provide some insight.

One of the less positive takeaways from this particular test case was the lack of consistency demonstrated when there was no experimental data with which to compare. This is evident in the spread in computational results between Test Case 3.2 in Figure 21 and Test Case 2.4 in Figure 16. Using two very comparable geometries, CFD results computed for Test Case 3.2, where no experimental data was available, demonstrated larger differences between participants than in Test Case 2.4, where experimental data was provided. From this observation, it may appear that there is no general consensus on modeling or gridding practices that are suitable for blind prediction. Because of a smaller number of samples available for this test case, there is clearly more work required in this area.

Lastly, despite the large spread in results, when comparing the predicted trends in Reynolds number from participant to participant, there was reasonably good agreement with expectations in terms of lift at angle of attack. This was shown in Figure 24, and discussed further in Section VI.

VIII. Summary

The 5th High-Lift Prediction Workshop continued the use of TFGs and collaborative meetings over the course of more than a year prior to the workshop, to accelerate the maturation of computational methods to predict high-lift aerodynamics. One key addition to the logistics of this workshop was performing the Test Cases one at a time, such that all TFGs were focused on a single test case for a longer duration of time, then holding focused mini-workshops to collaboratively discuss results for that test case across TFGs before moving onto the next test case. Collaboration in this manner was found to be very successful, and enabled many issues and findings to be explored jointly.

Test Case 1 consisted of a single test point using the CRM-HL wing-body configuration, executed in conjunction with the AIAA High-Fidelity CFD Verification Workshop, to explore grid convergence and verification for RANS solvers using a single turbulence model. It also served as a challenging case to demonstrate grid convergence and explore solver settings for methodologies beyond RANS. Importantly, many RANS participants were able to demonstrate verification on this problem using a common turbulence model. Other TFGs were less consistent, hampered by issues such as boundary layer transition, which prevented the HRLES and WMLES TFGs from achieving consensus on a consistent solution.

The second test case for HLPW-5 focused on model build up, as envisioned for the original model architecture of the CRM-HL. This test case consisted of four geometric configurations of increasing complexity, with the computation of full lift polars required for each. Although the computational resource requirements were large, most participants were able to complete the full request. Across TFGs, participants struggled in particular with predicting the flow features associated with the addition of the trailing-edge flap, highlighting an area where future workshops should focus. Results were sensitive to boundary layer transition on the leading-edge elements, suggesting the need for additional experimental data to better characterize the boundary layer flow and enable further refinement of predictive capability. As with previous workshops, RANS methodologies failed to consistently predict maximum lift, but are fairly consistent in the linear range of the lift versus angle of attack relationship. Adaptive RANS was able to demonstrate consistent grid convergence in the linear range using fewer points than required for hand-built grids. Scale resolving methods including WMLES and HRLES were more consistent at predicting maximum lift, but that improved accuracy comes at a drastically increased computational cost.

Test Case 3 explored Reynolds number trends from 1 million up to 30 million chord Reynolds number. This test case was intended to be compared to data collected in the NASA NTF, but inconsistencies found between the experimental dataset and other CRM-HL experimental results led to not having the NTF experimental data available for HLPW-5. Despite this, several WMLES participants produced results suggesting that grid convergence may be achievable, even at the highest Reynolds numbers. Both RANS and WMLES participants were able to produce lift versus Reynolds number trends in line with expectations.

There was limited participation from the High-Order TFG. These methodologies continue to hold promise, particularly in reducing computational cost for scale resolving methods. However, maturation of these technologies continue to lag other approaches.

Finally, one takeaway from earlier workshops was the necessity to run in-tunnel simulations for proper validation. However, this workshop largely deviated from that guidance, both because Test Case 2 was a full model and tunnel

effects from the ONERA test were much smaller than in previous wind-tunnel tests, but also because the large number of CFD simulations requested for this workshop were overwhelming for most participants. It is expected that future workshops will revisit a subset of these test cases, and dive deeper into topics such as running in tunnel, and boundary layer transition.

Acknowledgments

A. Placeholder

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